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AGE OF INNOVATION

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TEAM NAME: CODERS CREED

PROJECT NAME:
NEERKAVALAN (GUARDIAN OF WATER BODIES)





TRACK : MAKERFAIR

SDG FOCUS : CLIMATE ACTION & LIFE ON LAND

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RAJALAKSHMI ENGINEERING COLLEGE

TEAM MEMBERS

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III ECE
(TEAM LEAD & SYSTEM DESIGN)



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II ECE
(HARDWARE DEVELOPMENT)

Problem Statement

- Water bodies like lakes, ponds, and canals are gradually disappearing due to illegal encroachments and waste dumping.
- These issues reduce water storage and increase the risk of floods and water scarcity.
- Most encroachments are noticed only after serious damage has already occurred.
- Existing monitoring methods are manual, irregular, and not reliable.
- Complaints often lack proper evidence, leading to delayed or no action.



Core Problem:

- There is no continuous and reliable system to monitor, detect, and ensure action against encroachments.

Introduction:

- Water is one of the most essential resources for life, agriculture, and ecosystems.
- Protecting water bodies is becoming difficult due to increasing human activities.
- Technology can help monitor and protect these resources more effectively.
- Many water bodies are shrinking due to encroachments, pollution, and unplanned development.
- This leads to flooding, environmental damage, and loss of biodiversity.
- Continuous monitoring is needed instead of occasional inspections.
- A **smart system** can help in early detection and protection of water bodies.

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RESEARCH SURVEY:

CURRENT SYSTEMS DEPEND MOSTLY ON MANUAL INSPECTION.

🔗 REFERENCE:URBAN WATER BODY MONITORING CHALLENGES (MANUAL DEPENDENCY)

SATELLITE MONITORING IS NOT FREQUENT ENOUGH FOR EARLY DETECTION.

🔗 REFERENCE:LIMITATIONS OF SATELLITE TEMPORAL RESOLUTION FOR ENVIRONMENTAL MONITORING

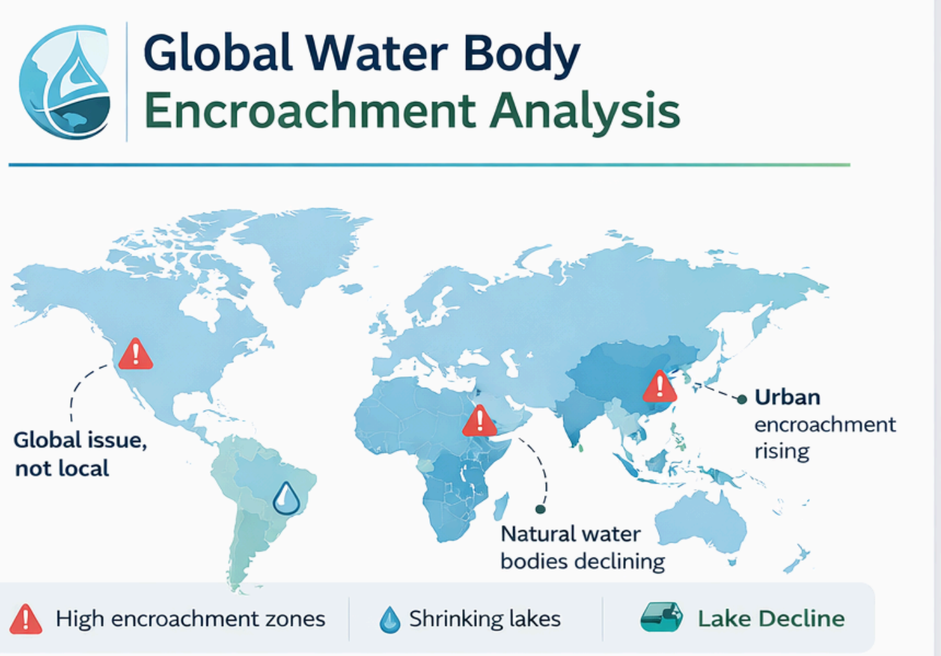
NO PROPER INTEGRATION BETWEEN MONITORING AND REPORTING SYSTEMS.

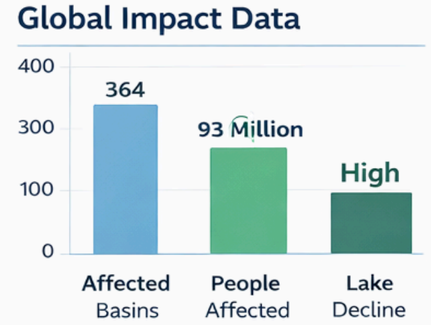
🔗 REFERENCE:SMART WATER MANAGEMENT SYSTEM GAPS AND INTEGRATION CHALLENGES

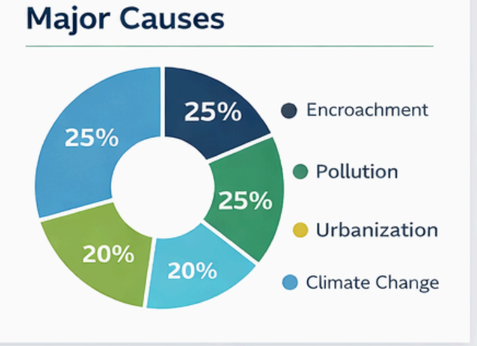
LACK OF STRONG PROOF LEADS TO WEAK ENFORCEMENT.

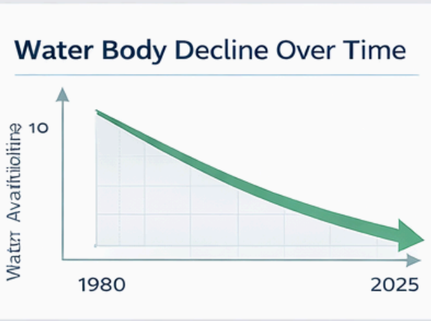
🔗 REFERENCE:WATER BODY ENCROACHMENT AND GOVERNANCE ISSUES IN INDIA

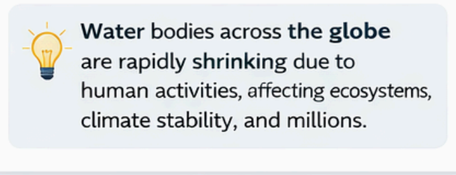
OBSERVATION:
EXISTING SOLUTIONS REACT AFTER DAMAGE INSTEAD OF PREVENTING IT











NEERKAVALAN

GUARDIAN OF WATER BODIES

NEER — WATER

KAVALAN — GUARDIAN

PROPOSED SOLUTION

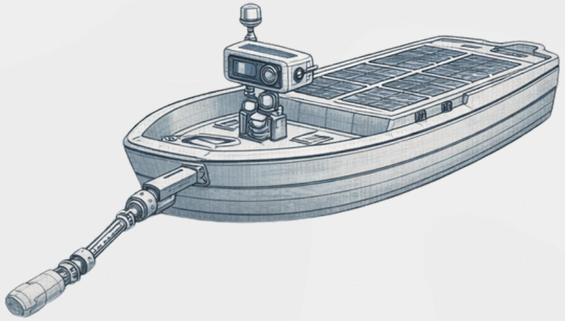
NeerKavalan is a dual-system solution:

- 🚁 **Hexacopter Drone (Aerial Monitoring)**
- 🚤 **Water Rover (Surface Inspection)**

HOW IT ADDRESSES THE PROBLEM

- Analyses collected data
- Identifies changes and encroachments
- Generates proof-based reports
- AI-Powered Safe Navigation
- Coordinated Relief Management





DRONE ROLE:

- Monitors large areas from above
- Captures images with GPS and time data
- Detects encroachments using AI

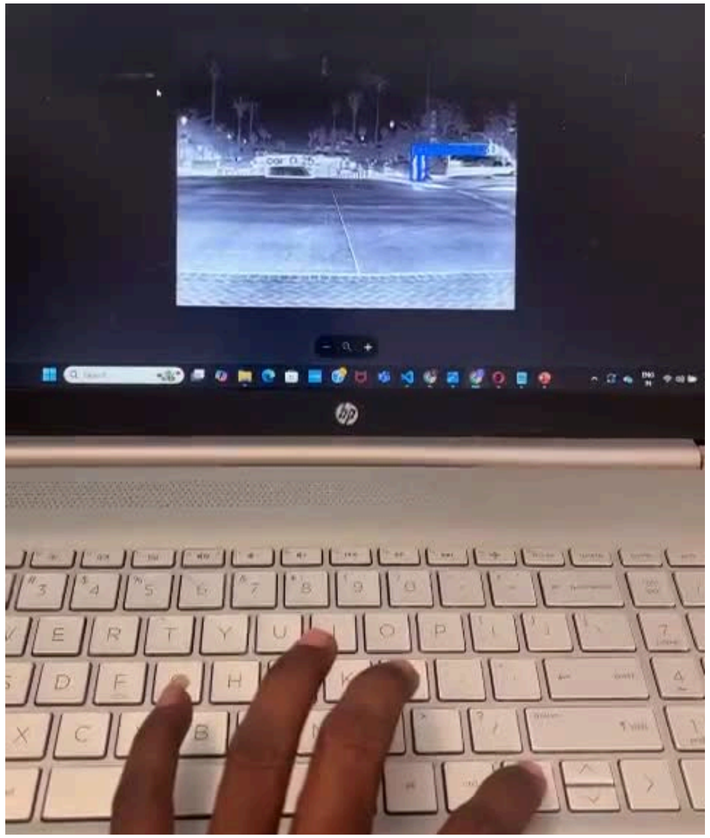
WATER ROVER ROLE:

- Moves on water surface
- Performs close inspection
- Detects dumping, pollution, and boundary violations

A TWO-LEVEL MONITORING SYSTEM THAT ENSURES BOTH DETECTION AND VERIFICATION.

Prototype (if applicable):







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* ADVANTAGES *

* DISADVANTAGES *



ADVANTAGES	DISADVANTAGES
Covers large areas quickly	Weather-dependent operation
Provides strong multi-level evidence	dependence on AI and sensors
Real time data transmission and security	potential signal loss Initial setup cost
Faster and more efficient Dual system increase	Requires maintenances





References:

HTTPS://WWW.FEEDGLASS.COM/TOP/DRONES-BLOGS?UTM_SOURCE=CHATGPT.COM

HTTPS://WWW.FINANCIALEXPRESS.COM/LIFE/TECHNOLOGY-MAHINDRA-GROUP-AND-SENTRYCS-TO-DEVELOP-ANTI-DRONE-SOLUTIONS-3597618/?UTM_SOURCE=CHATGPT.COM

HTTPS://WWW.DIGIKEY.COM/?UTM_SOURCE=CHATGPT.COM

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Thank You

